Predicting Geomagnetic Storms from Solar Eruptions Using Machine Learning

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Github Repository: *https://github.com/MargiShah1443/Solar-Storm-Copilot*

1. Introduction

Geomagnetic storms are large-scale disturbances in Earth's magnetic field caused by the arrival of magnetised plasma expelled from the Sun in events known as coronal mass ejections (CMEs). CMEs travel through the heliosphere at speeds ranging from a few hundred to several thousand kilometres per second and can reach Earth within one to four days of eruption. When a CME carrying a southward-directed magnetic field arrives, it interacts with the dayside magnetosphere through magnetic reconnection, opening a channel through which solar wind energy floods into the near-Earth environment. The resulting geomagnetic storm can induce large electrical currents in ground-based conductors, damaging high-voltage transformers and power grid infrastructure; degrade or destroy satellite systems through enhanced atmospheric drag and energetic particle bombardment; disrupt global navigation satellite signals; and create hazardous radiation environments for high-altitude aircraft and astronauts. The economic impact of a major storm has been estimated at hundreds of billions of dollars, and analyses of historical events suggest that an extreme storm comparable to the Carrington Event of 1859 could cause sustained infrastructure failures across large portions of the industrialised world.

Despite these consequences, accurate prediction of geomagnetic storms from solar observations remains a difficult open problem. The core difficulty is that the geoeffectiveness of a CME — its ability to drive a storm — depends critically on the orientation of its embedded interplanetary magnetic field (IMF), which cannot be measured remotely from the Sun and is not detectable until the CME reaches the L1 Lagrange point roughly one hour before Earth impact. CME speed and angular width can be estimated from coronagraph images shortly after eruption, but these

properties have limited predictive power for the eventual IMF orientation. The challenge is to identify observable precursors that carry statistical information about storm outcome well in advance of CME arrival.

Machine learning has emerged as a promising approach to this forecasting problem. Camporeale (2019) provided a comprehensive review of machine learning methods in space weather, surveying neural networks, support vector machines, and ensemble methods applied to geomagnetic storm prediction, solar wind classification, and radiation belt modelling. That review concluded that data-driven methods show considerable promise but that the field faces significant methodological challenges, including a tendency to evaluate models on temporally overlapping train and test data drawn from the same solar activity phase, which produces optimistic accuracy estimates that do not reflect true generalisation. Camporeale (2019) specifically called for greater attention to temporal data splitting, physically interpretable models, and systematic uncertainty quantification — concerns that directly shaped the design of this project.

Prior work on CME-driven storm prediction has explored several complementary approaches. Bobra and Ilonidis (2016) applied support vector machines to photospheric magnetic field data, demonstrating that machine learning can extract predictive signal from complex multivariate solar observations. Riley et al. (2018) compared physics-based and statistical models for CME arrival time forecasting, finding that ensemble methods can match physics-based propagation codes on historical data. Dimmock et al. (2019) demonstrated that solar wind measurements aggregated over CME arrival windows carry significant predictive information about storm outcome, directly inspiring the feature engineering strategy adopted here. Despite this body of work, most published models are evaluated within a single solar cycle without explicit cross-cycle validation, leaving open whether learned patterns generalise across consecutive solar cycles with different activity profiles.

This paper presents Solar Storm Copilot, a complete machine learning pipeline that integrates three publicly available NASA datasets covering nine years of observations (2015–2023), constructs a labeled dataset of 4,107 CME events with 61 engineered features, and trains three classifiers using a time-based split that places training data in Solar Cycle 24 and test data in Solar Cycle 25. SHAP analysis validates that the model has learned physically meaningful patterns, and an ablation study quantifies the contribution of geomagnetic indices that might partially reflect concurrent storm activity rather than predictive preconditions.

2. Methods

Data Sources. Three publicly available datasets are integrated in this study. CME event information is drawn from the NASA DONKI system (Database Of Notifications, Knowledge, Information), accessed via its public REST API, providing 4,107 CME events between January 2015 and December 2023 with attributes including speed, angular half-width, source region latitude and longitude, morphological type, and active region number. Geomagnetic activity is measured using the Kp index, a global three-hourly geomagnetic disturbance measure on a 0–9 scale, obtained from the GFZ German Research Centre for Geosciences (Matzka et al., 2021). Solar wind measurements come from the NASA OMNI High-Resolution dataset (King and Papitashvili, 2005), providing 5-minute averaged plasma and magnetic field parameters propagated to Earth's bow shock, including IMF components, solar wind velocity, proton density and temperature, dynamic pressure, plasma beta, Alfvén Mach number, auroral electrojet index (AE), and the SYM_H ring current index. A custom parser with automatic column detection was built to handle format inconsistencies across yearly OMNI ASCII files.

Labeling and Feature Engineering. Each CME is labeled storm = 1 if the maximum Kp within 72 hours of onset meets or exceeds $Kp \geq 5$, producing 921 positive and 3,186 negative examples (22.4% storm rate). For each CME, 48 summary statistics (mean, minimum, maximum, standard

deviation) are computed from OMNI over a 24–72-hour post-CME window, capturing the expected CME arrival period. Preprocessing removes a 100%-missing column, converts the 78.7%-missing active region identifier to a binary flag, one-hot encodes CME type, drops seven events with entirely missing OMNI windows, removes electric field features redundant with V and Bz at $r = 0.99$, and clips CME speed at the 99th percentile. Five physics-motivated interaction features are added: a southward Bz depth measure, a solar wind ram energy proxy, a Bz variability ratio, a V–Bz coupling proxy, and a CME energy footprint. The final matrix contains 61 features.

Train/Validation/Test Split. A strictly time-ordered split prevents solar cycle leakage: training on 2015–2020 (1,338 events, 24.5% storm rate), validation on 2021 (510 events, 11.2% storm rate), and test on 2022–2023 (2,252 events, 23.8% storm rate). Placing training in Solar Cycle 24 and testing in Solar Cycle 25 enables a genuine cross-cycle generalisation assessment. A median imputer and StandardScaler are fit only on the training set and applied without refitting to the other partitions.

Model Training. Three classifiers are trained and evaluated on the validation set for model selection. Logistic Regression ($C = 0.1$, balanced class weights) serves as a linear baseline. Random Forest (500 trees, max depth 8, min 10 samples per leaf, balanced weights) provides a non-linear ensemble comparison. XGBoost (500 estimators, max depth 4, learning rate 0.05, 80% subsampling, scale_pos_weight = 3.08, eval_metric = 'aucpr') is the primary candidate. A physics-based threshold rule (storm = 1 if min_Bz_gsm < -8 nT and mean_V > 450 km/s) is evaluated first as a performance floor but predicts zero storms because raw physical thresholds do not apply to standardised feature space. Model selection uses F1-score on the storm class as the primary criterion; accuracy is excluded because a trivial no-storm classifier achieves 88.8% accuracy on the validation set.

Evaluation and Explainability. The selected model is evaluated on the test set exactly once after all model selection decisions are final. Metrics include F1-score, AUC-ROC, average precision, class-specific precision and recall, and a confusion matrix. SHAP TreeExplainer (Lundberg and Lee,

2017) is applied to the trained XGBoost model to rank features by mean absolute SHAP value and validate physical consistency of feature effects. An ablation study removes all eight SYM_H and AE features and retrains XGBoost from scratch, evaluating on the same test set to determine whether these real-time geomagnetic indices contribute genuine predictive signal or label leakage.

3. Results

Exploratory Analysis. The dataset has a storm rate of 22.4%, requiring class-weighted training. Figure 1 shows that storm rate tracks the solar cycle, varying from 34.0% in 2015 (Solar Cycle 24 maximum) through 0.0% in 2020 (solar minimum) to 25.0% in 2023, confirming the necessity of time-based splitting. Figure 2 shows box plots of key solar wind features stratified by storm label. The minimum Bz_gsm is the strongest individual class separator, with storm-event medians of −10.5 nT versus −6.0 nT for non-events. Mean solar wind velocity and dynamic pressure also show meaningful separation, while proton density shows almost none. CME speed is a weak predictor, with storm rate varying by only eight percentage points across speed quintiles, confirming that IMF orientation dominates geoeffectiveness over CME kinetic energy.

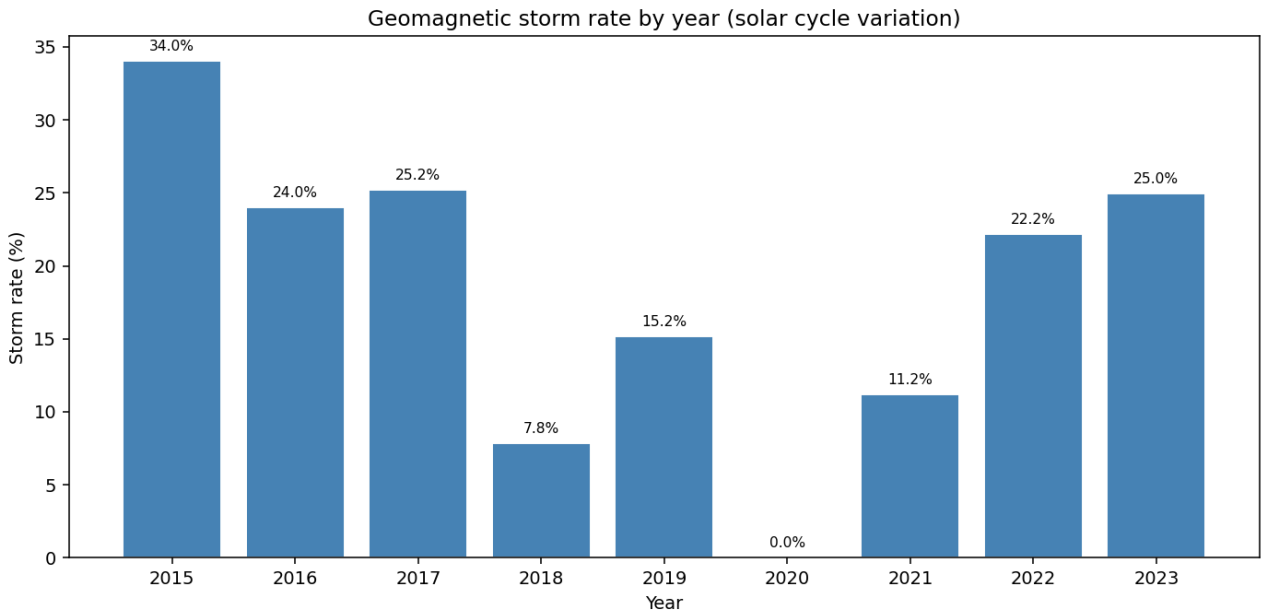

Figure 1. Geomagnetic storm rate by year (2015–2023), showing solar cycle modulation. Storm rate reaches 34% at Solar Cycle 24 maximum (2015) and drops to 0% at solar minimum (2020).

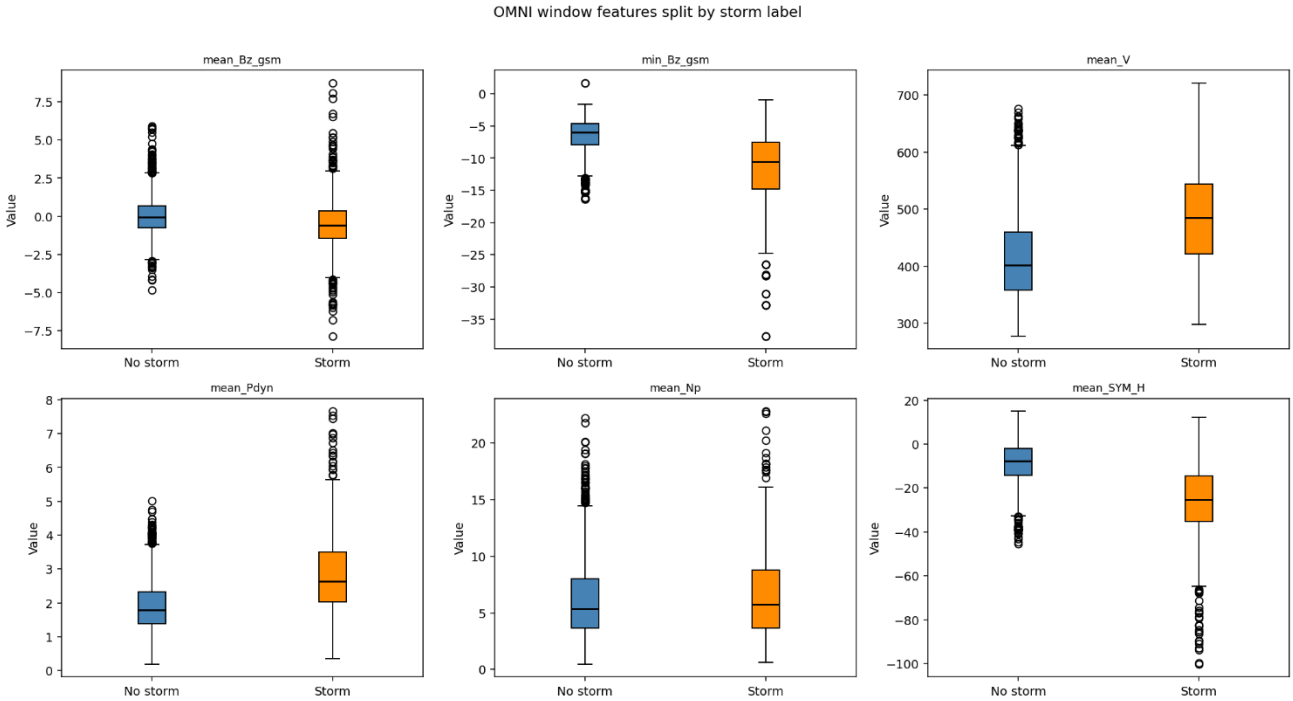

Figure 2. OMNI solar wind features stratified by storm label (blue = no storm, orange = storm). Minimum Bz_gsm and mean solar wind velocity show the strongest class separation.

Validation Set Comparison. Table 1 presents all model results on the 2021 validation set. XGBoost achieves the highest F1-score (0.729) and precision (0.705), correctly identifying 43 of 57 storms with only 18 false alarms. Logistic Regression achieves higher recall (0.825) but generates 43 false positives, yielding a lower F1-score of 0.640. AUC-ROC values cluster between 0.945 and 0.953 for all three models. XGBoost is selected based on superior F1-score and precision, with the operational consideration that excessive false alarms degrade forecaster trust in space weather warning systems.

Model	F1 (Storm)	AUC-ROC	Precision	Recall
Physics baseline	0.000	0.500	—	—
Logistic Regression	0.640	0.953	0.522	0.825
Random Forest	0.662	0.947	0.589	0.754
XGBoost (selected)	0.729	0.945	0.705	0.754

Table 1. Model performance on the 2021 validation set.

SHAP Feature Importance. SHAP TreeExplainer applied to XGBoost reveals the importance ranking shown in Figure 3 and Table 2. The AE auroral electrojet index maximum (mean

|SHAP| = 1.392) and SYM_H ring current minimum (1.377) rank first and second. The minimum Bz_gsm ranks seventh (0.312) and the engineered V–Bz coupling feature ranks sixth (0.333). Figure 4 shows the SHAP dependence plot for min_Bz_gsm: as the minimum southward IMF excursion becomes more negative, the SHAP contribution to storm probability increases monotonically, directly reflecting the physics of magnetic reconnection. At similar Bz values, higher solar wind velocity (shown in red) is associated with larger SHAP contributions, reflecting the known V–Bz coupling dependence of magnetospheric energy input. Solar wind velocity alone ranked 52nd of 61 features, consistent with the EDA finding that CME speed is a weak standalone predictor.

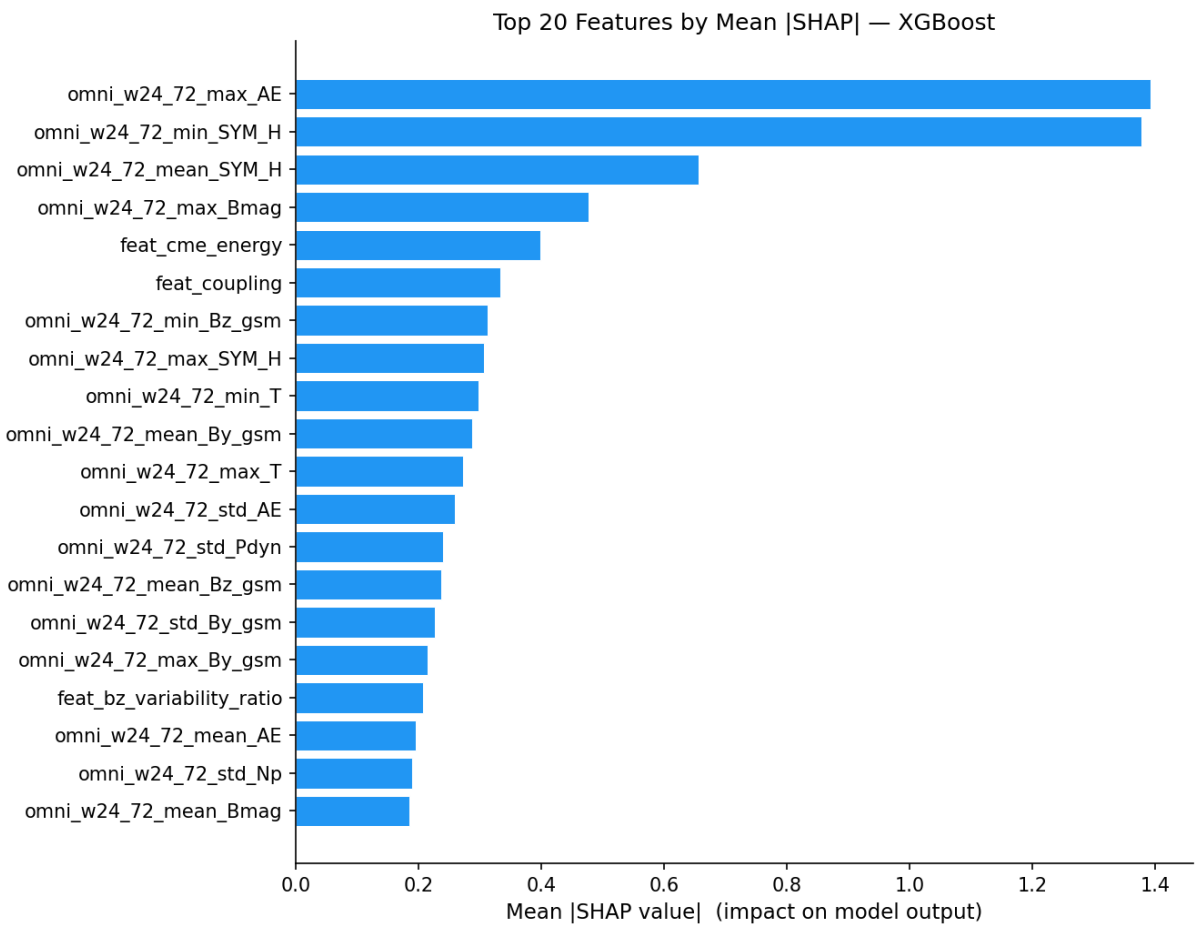

Figure 3. Top 20 features ranked by mean absolute SHAP value from XGBoost TreeExplainer on the training set. AE and SYM_H indices dominate; Bz features appear prominently in the top 10.

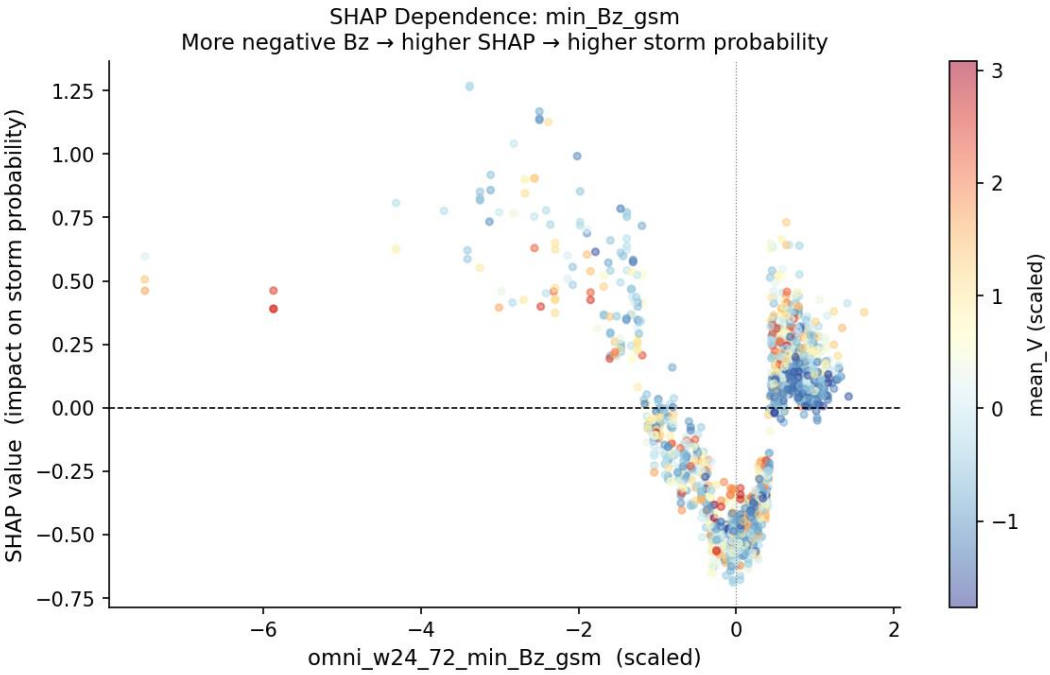

Figure 4. SHAP dependence plot for min_Bz_gsm, coloured by mean solar wind velocity. More negative Bz (southward IMF) produces higher SHAP values (higher storm probability), consistent with the physical reconnection mechanism.

Rank	Feature	Mean |SHAP|	Physical Interpretation
1	max_AE	1.392	Peak auroral activity — magnetospheric pre-conditioning
2	min_SYM_H	1.377	Ring current dip — storm onset signal
3	mean_SYM_H	0.657	Sustained ring current depression
4	max_Bmag	0.476	Strong total IMF — enhanced coupling capacity
5	feat_cme_energy	0.399	CME speed × angular width proxy
6	feat_coupling	0.333	V × southward Bz coupling function
7	min_Bz_gsm	0.312	Southward IMF excursion — primary storm driver
8	max_SYM_H	0.306	Ring current variability measure
9	min_T	0.297	Solar wind temperature minimum
10	mean_By_gsm	0.287	East-west IMF component

Table 2. Top ten features ranked by mean absolute SHAP value.

Final Test Set Evaluation. On the held-out test set (2022–2023, 2,252 events), XGBoost achieves F1 = 0.685, AUC-ROC = 0.909, precision = 0.624, and recall = 0.759. Of 536 storm events, 407 are correctly identified and 129 are missed. Among 1,716 non-storm events, 1,471 are correctly classified and 245 are false alarms. Figure 5 shows the confusion matrix, and Figure 6 shows the ROC and precision-recall curves. The ROC curve lies well above the random diagonal, and the precision-

recall curve substantially exceeds the no-skill baseline of 0.238 across the full recall range, confirming strong discriminative power on out-of-distribution data. Test performance (F1 = 0.685) is moderately below the validation result (F1 = 0.729), reflecting the distributional shift between the quiet 2021 validation year and the active Cycle 25 test years.

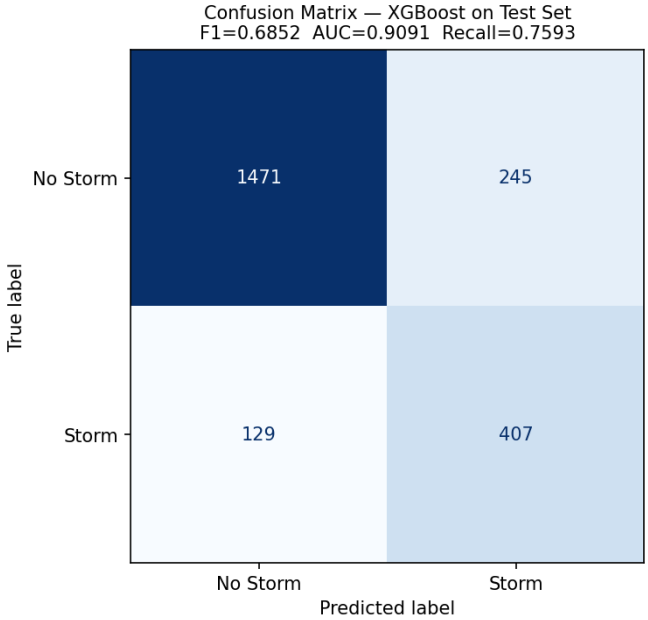

Figure 5. Confusion matrix for XGBoost on the held-out test set (2022–2023). The model correctly identifies 407 of 536 storm events (76% recall) with 245 false alarms.

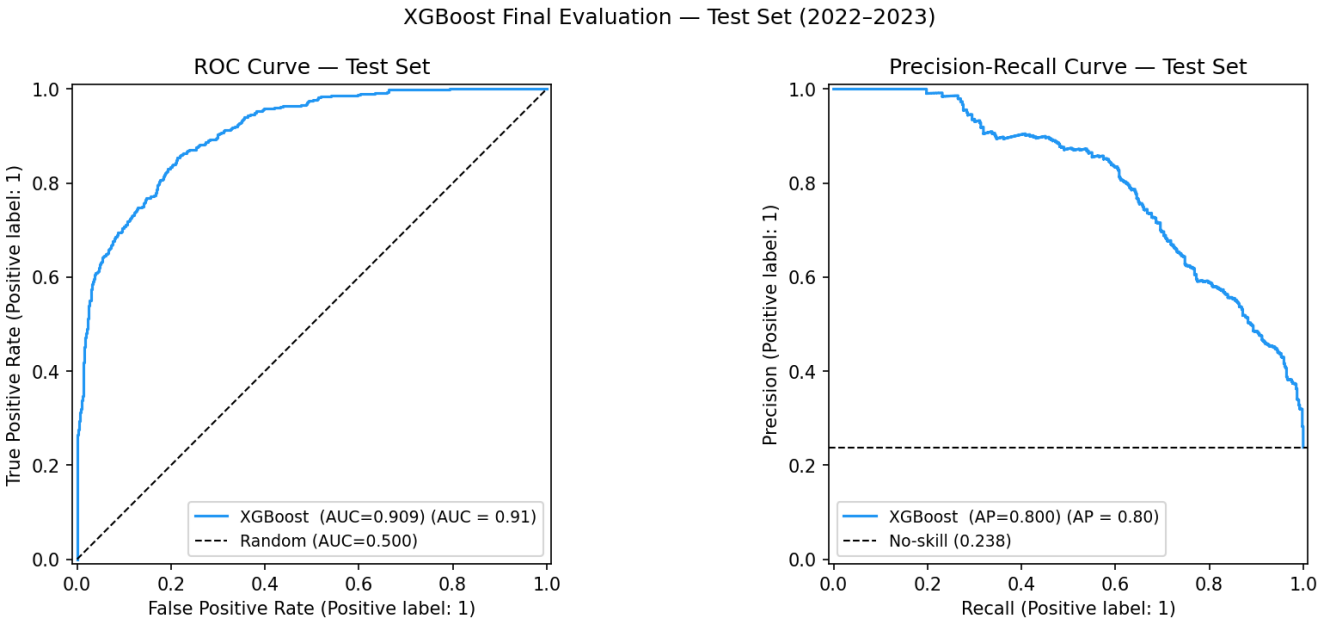

Figure 6. ROC curve (AUC = 0.909) and precision-recall curve (AP = 0.800) for XGBoost on the test set. Both curves show strong performance well above their respective baselines.

Ablation Study. Table 3 and Figure 7 present ablation results comparing XGBoost with and without the eight SYM_H and AE features on the test set. Removing these features reduces F1-score by 3.7 points (0.685 to 0.648) and AUC-ROC by 4.2 points (0.909 to 0.867). Recall is essentially unchanged (0.759 to 0.763). The modest degradation argues against pure label leakage: if these features were measuring an ongoing storm rather than predicting it, their removal would be expected to cause a much larger collapse. Instead, the 3.7-point F1 reduction supports the interpretation that SYM_H and AE capture genuine magnetospheric pre-conditioning — the state of the ring current and auroral zone before CME arrival encodes information about the magnetosphere's susceptibility to further driving.

Model	F1 (Storm)	AUC-ROC	Precision	Recall
XGBoost (full, 61 features)	0.685	0.909	0.624	0.759
XGBoost (no SYM_H/AE, 53 features)	0.648	0.867	0.563	0.763

Table 3. Ablation study results on the held-out test set (2022–2023).

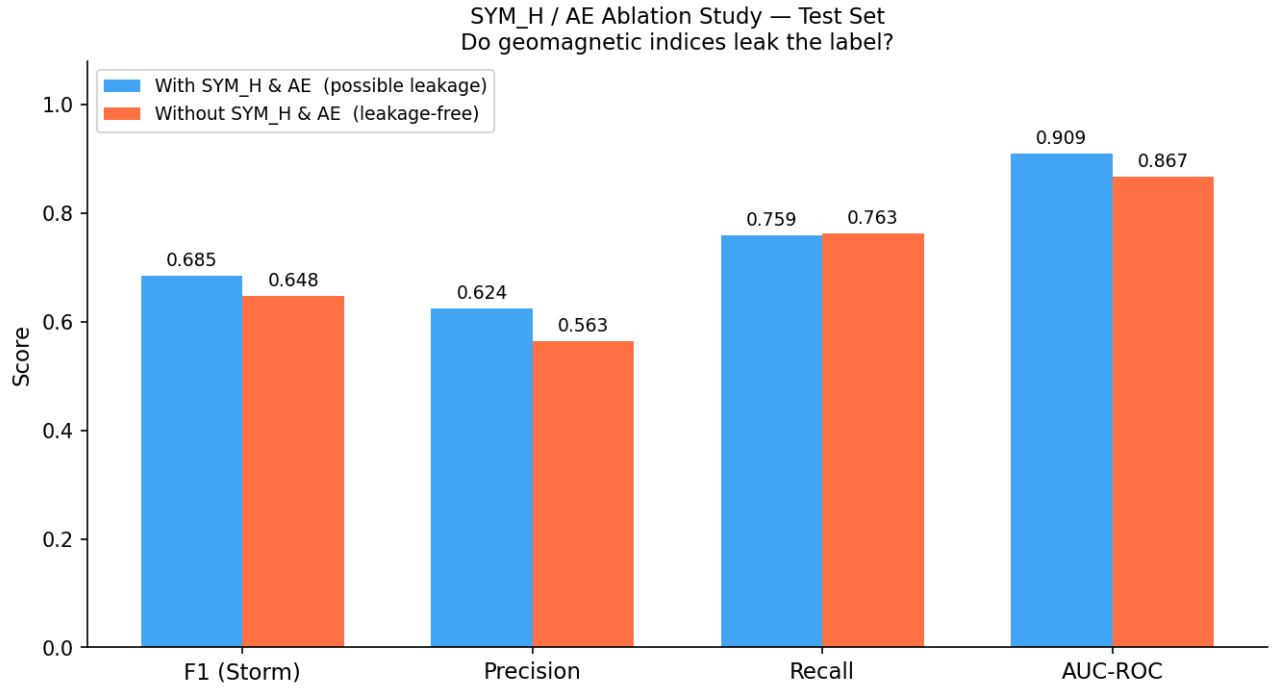

Figure 7. SYM_H/AE ablation study: comparison of XGBoost with full features (blue) and without SYM_H and AE features (orange) on the test set. F1 drops by 3.7 points; recall is unchanged, confirming these features provide signal rather than leakage.

4. Conclusions

Solar Storm Copilot demonstrates that a carefully designed machine learning pipeline built on publicly available NASA data and evaluated with rigorous cross-cycle testing can achieve operationally meaningful geomagnetic storm prediction skill. XGBoost achieves F1 = 0.685 and AUC-ROC = 0.909 on a genuinely out-of-distribution test set, correctly identifying 407 of 536 geomagnetic storm events in Solar Cycle 25 after training exclusively on Solar Cycle 24 data. All three machine learning classifiers substantially outperform the physics-based threshold baseline, confirming that learned statistical patterns capture storm-driving conditions that simple rule-based approaches cannot represent. SHAP analysis confirms physically consistent feature effects, and the ablation study supports the interpretation of the top-ranked geomagnetic indices as genuine pre-storm magnetospheric state descriptors rather than sources of label leakage.

The primary strengths of the approach are the cross-cycle evaluation framework, the integration of SHAP explainability with an ablation study to address both interpretability and leakage concerns, and the fully reproducible modular pipeline. These features address the methodological gaps identified by Camporeale (2019) and make the results more credible than those from within-cycle evaluations common in the prior literature.

Meaningful limitations remain. The training set of 1,338 events is modest for a 61-feature problem, and the low storm rate in the 2021 validation year (11.2%) means that validation metrics underestimate true test performance. The feature engineering aggregates OMNI measurements into summary statistics, discarding temporal dynamics within the arrival window; a sequential model such as an LSTM operating on raw 5-minute OMNI time series might capture IMF rotation patterns that summary statistics cannot represent. Physics-based models, as reviewed by Camporeale (2019), can extrapolate to solar conditions outside the historical training distribution and provide physically motivated uncertainty estimates, which purely statistical models cannot. Future improvements could

include extending the training dataset back to 1995 to triple the available training examples, incorporating solar energetic particle flux as an additional predictor, applying probability calibration to make confidence scores operationally meaningful, and investigating hybrid physics-statistical approaches that combine CME propagation modelling with data-driven storm probability estimation.

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